

PROJECT REPORT DMS613

Option Valuation Using Fast Fourier Transform

1 Intended Outcomes

- Understanding the derivation of the Fourier transform of a modified call price and exploring why the damping coefficient α is necessary to ensure square integrability of call pricing function.
- To study how the analytic expression for the Fourier transform is mapped onto the FFT algorithm to achieve real-time pricing.
- To study VG processes and its role in addressing the biases of standard Black Scholes Model and how is FFT implemented in VG Processes.

2 Why need for the Fast Fourier Transform ?

- The pricing of a European Call Option involves calculating the expected value of payoff under a risk-neutral probability measure. It requires integrating the Probability Distribution Function (PDF).
- However, in many advanced models like the Variance Gamma Model have complex PDF but they often have simpler Characteristic Function(ϕ) that is the Fourier transform of the PDF.
- It is computationally more efficient and reduces the time complexity to $O(N \log N)$ from $O(N^2)$.

3 The Fourier Approach For Option Pricing

3.1 The Modified Transform

- $s_T = \ln(S_T)$ is the log-price of the asset at maturity T and $q_T(s)$ is its risk-neutral density. $q_T(s)$ be its risk-neutral density. Then the value of a European call option with log-strike price $k = \ln(K)$ is:

$$C_T(k) = \int_k^\infty e^{-rT} (e^s - e^k) q_T(s) ds$$

- As $k \rightarrow -\infty$, $C_T(k) \rightarrow S_0$, making the function non-square integrable and thus Fourier Transform cannot be applied on $C_T(k)$.

- The author in the paper defined a modified call price $c_T(k) \equiv e^{\alpha k} C_T(k)$ for $\alpha > 0$, the function is square integrable.
- The Fourier transform of $c_T(k)$ is:

$$\psi_T(v) = \int_{-\infty}^{\infty} e^{ivk} c_T(k) dk = \int_{-\infty}^{\infty} e^{ivk} \int_k^{\infty} e^{\alpha k} e^{-rT} (e^s - e^k) q_T(s) ds$$

- Now after swapping the order of integration $\psi_T(v)$ becomes:

$$\psi_T(v) = \int_{-\infty}^{\infty} e^{-rT} q_T(s) \int_{-\infty}^s (e^{s+\alpha k+ivk} - e^{(\alpha+1)k+ivk}) dk ds$$

- Evaluating the inner integral with respect to k gives:

$$\begin{aligned} \int_{-\infty}^s (e^s e^{(\alpha+iv)k} - e^{(\alpha+1+iv)k}) dk &= \left[\frac{e^s e^{(\alpha+iv)k}}{\alpha + iv} - \frac{e^{(\alpha+1+iv)k}}{\alpha + 1 + iv} \right]_{-\infty}^s \\ &= \frac{e^{(\alpha+1+iv)s}}{\alpha + iv} - \frac{e^{(\alpha+1+iv)s}}{\alpha + 1 + iv} = \frac{e^{(\alpha+1+iv)s}}{(\alpha + iv)(\alpha + 1 + iv)} \end{aligned}$$

- Evaluating the outer integral now:

$$\psi_T(v) = \frac{e^{-rT}}{(\alpha + iv)(\alpha + 1 + iv)} \int_{-\infty}^{\infty} e^{i(v-(\alpha+1)s)} q_T(s) ds$$

- Since, the characteristic function $\phi_T(u) = E[e^{iusT}]$, the integral finally simplifies into the expression:

$$\psi_T(v) = \frac{e^{-rT} \phi_T(v - (\alpha + 1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v}$$

3.2 Selection Of Damping Constant (α)

- The selection of the damping coefficient α is governed by the requirement that the modified call price $c_T(k)$ must be square-integrable, which implies that the transform $\psi_T(v)$ must be well-defined at the origin ($v = 0$).
- The value of the transform $\psi_T(v)$ at $v = 0$ is:

$$\psi_T(0) = \frac{e^{-rT} E[S_T^{\alpha+1}]}{\alpha(\alpha + 1)}$$

- For $\psi_T(0)$ to be finite the final bound on α is :

$$E[S_T^{\alpha+1}] < \infty$$

3.3 Truncation Error Analysis

- To implement the pricing formula numerically via the FFT, the semi-infinite integral must be truncated at an upper limit a .
- The truncation error ϵ is defined as the contribution of the integrand over the omitted interval $[a, \infty)$. Mathematically, it is expressed as:

$$\epsilon(a) = \frac{e^{-\alpha k}}{\pi} \int_a^\infty \text{Re}[e^{-ivk} \psi_T(v)] dv$$

- Taking modulus both the sides:

$$|\epsilon(a)| \leq \frac{e^{-\alpha k}}{\pi} \int_a^\infty |\psi_T(v)| dv$$

- Rewriting the transform:

$$\psi_T(v) = \frac{e^{-rT} \phi_T(v - (\alpha + 1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v}$$

- The numerator is bounded by $E[S_T^{\alpha+1}]$, which is a constant A independent of v and in the denominator the $-v^2$ term dominates, thus we can say that the modulus of the transform decays quadratically as:

$$|\psi_T(v)| \leq \frac{A}{v^2}$$

- Finally substituting the bound back into the error integral:

$$|\epsilon(a)| \leq \frac{e^{-\alpha k}}{\pi} \int_a^\infty \frac{A}{v^2} dv = \frac{Ae^{-\alpha k}}{\pi} \left[-\frac{1}{v} \right]_a^\infty = \frac{Ae^{-\alpha k}}{\pi a}$$

4 Fourier approach for pricing Out-of-the-Money Options

- While the damped call price method works well for most maturities but it encounters numerical instability for options with very short maturities.
- As the time to maturity T vanishes, the call price function $C_T(k)$ converges to the payoff function $(S_0 - e^k)^+$.
- This function possesses a kink at $S_0 = e^k$. In the Fourier domain, this discontinuity in the first derivative results in an integrand that oscillates at high frequencies which produces significant noise.

- To smoothen the integrand, we focus on the time Value $z_T(k)$ which is defined as the option price minus its lower bound (intrinsic value). For a call option, it can be written as:

$$z_T(k) = C_T(k) - (S_0 e^{-rT} - e^k)^+$$

- By substituting the expression for the intrinsic value, the paper provides the explicit form for $z_T(k)$ as:

$$z_T(k) = e^{-rT} \int_{-\infty}^{\infty} [(e^k - e^s)1_{s < k, k < 0} + (e^s - e^k)1_{s > k, k > 0}] q_T(s) ds$$

- The Fourier Transform of $z_T(k)$ is:

$$\zeta_T(v) = \int_{-\infty}^0 e^{ivk} e^{-rT} \int_{-\infty}^k (e^k - e^s) q_T(s) ds, dk + \int_0^{\infty} e^{ivk} e^{-rT} \int_k^{\infty} (e^s - e^k) q_T(s) ds, dk$$

- By swapping the order of integration and simplifying we get the expression for $\zeta_T(v)$ as:

$$\zeta_T(v) = e^{-rT} \left(\frac{1}{1 + iv} - \frac{e^{rT}}{iv} - \frac{\phi_T(v - i)}{v^2 - iv} \right)$$

- Although there is no issue regarding the behavior of $z_T(k)$ as k tends to positive or negative infinity but as $T \rightarrow 0$, $z_T(k)$ (that is for very short duration options), $\zeta_T(v)$ behaves like a Dirac delta function at $k = 0$, causing the transform to become wide and oscillatory.
- To overcome this the author suggests using a hyperbolic sine damping function that ensures that the function vanishes at $k=0$.

$$\gamma_T(v) = \int_{-\infty}^{\infty} e^{ivk} \sinh(\alpha k) z_T(k) dk = \frac{\zeta_T(v - i\alpha) - \zeta_T(v + i\alpha)}{2}$$

- The final time value derived is thus:

$$z_T(k) = \frac{1}{\sinh(\alpha k)} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ivk} \gamma_T(v) dv$$

5 Option Pricing Using the Fast Fourier Transform

- The FFT computes the following sum for $k = 1, \dots, N$:

$$w(k) = \sum_{j=1}^N e^{-i \frac{2\pi}{N} (j-1)(k-1)} (j)$$

- To use the FFT algorithm we need to first of all discretize the continuous pricing integral,

$$C_T(k) = \frac{e^{-\alpha k}}{\pi} \int_0^a e^{-ivk} \psi_T(v) dv$$

We can use the Trapezoid Rule by setting $v_j = \eta(j-1)$. Now the call price approximately becomes:

$$C_T(k) \approx \frac{e^{-\alpha k}}{\pi} \sum_{j=1}^N e^{-iv_j k} \psi_T(v_j) \eta$$

and $a = N\eta$

- The authors employ a regular spacing of size λ for N values of k :

$$k_u = -b + \lambda(u-1)$$

for $u = 1, \dots, N$

- This provides a range from $-b$ to b , where $b = \frac{1}{2}N\lambda$. Substituting the grids for v_j and k_u into the approximation we get:

$$C_T(k_u) \approx \frac{e^{-\alpha k_u}}{\pi} \sum_{j=1}^N e^{-i\lambda\eta(j-1)(u-1)} e^{ibv_j} \psi_T(v_j) \eta$$

- To get the form required by the FFT summation, we must impose the following restriction on the grid step sizes:

$$\lambda\eta = \frac{2\pi}{N}$$

- This constraint implies a trade-off: a very small η leads to a large λ , which is not desirable.
- To overcome this the authors incorporate Simpson's rule weightings into our summation to get :

$$C(k_u) = \frac{e^{-\alpha k_u}}{\pi} \sum_{j=1}^N e^{-i\frac{2\pi}{N}(j-1)(u-1)} e^{ibv_j} \psi(v_j) \frac{\eta}{3} [3 + (-1)^j - \delta_{j-1}]$$

- For out of the money options the process is same we just need to replace $e^{(k\alpha)}$ by $\sinh(k\alpha)$.

6 Implementation On the Variance Gamma(VG) Process

- The VG process, $X_t(\sigma, \theta, \nu)$, is obtained by evaluating an arithmetic Brownian motion with drift θ and volatility σ at a random time given by a gamma process with a mean rate of 1 and a variance rate of ν .
- The risk-neutral process for the stock price is given by:

$$S_t = S_0 \exp[rt + X_t(\sigma, \theta, \nu) + \omega t]$$

where ω is the risk-neutral drift correction defined as:

$$\omega = \frac{1}{\nu} \ln \left(1 - \theta\nu - \frac{1}{2}\sigma^2\nu \right)$$

. This correction ensures that the mean rate of return on the stock equals the risk-free interest rate r .

- The primary advantage of the VG model is that while its density function is complex, its characteristic function for the log-price $s_T = \ln S_T$ is analytically simple given by:

$$\phi_T(u) = \exp[iu(\ln S_0 + (r + \omega)T)] \left(1 - i\theta\nu u + \frac{1}{2}\sigma^2 u^2 \nu \right)^{-T/\nu}$$

- Volatility smile refers to the empirical observation that at-the-money options have lower implied volatilities than in-the-money or out-of-the-money options, a phenomenon the Black-Scholes model cannot replicate.
- For the FFT to converge, the damping factor α must satisfy $E[S_T^{\alpha+1}] < \infty$. For the VG model, this requires:

$$\alpha < \sqrt{\frac{\theta^2}{\sigma^4} + \frac{2}{\sigma^2\nu}} - \frac{\theta}{\sigma^2} - 1$$

- In the paper the author states that setting $\alpha = 1.5$ is generally effective for the modified call price, while $\alpha = 1.10$ is often used for the modified time value approach for very short maturity call options.

7 Conclusions

- Through this project I successfully explored the Carr-Madan Fast Fourier Transform (FFT) framework for option valuation.

7.1 Key Observations

- The implementation of the FFT algorithm reduces the complexity of pricing an option chain from $O(N^2)$ to $O(N \log N)$.
- The use of a damping coefficient α is necessary to ensure the square-integrability of the call pricing function, effectively resolving the divergence issues.
- The VG process effectively removes the volatility smile bias of the standard Black-Scholes.
- The "Time Value" approach, utilizing sinh damping, provided a critical solution for short-dated options, suppressing the oscillations and maintaining precision as $T \rightarrow 0$.